Experiment No-26 : Fruit and Its Color using Backtracking

Aim:

To write a Prolog program to find a fruit and its color using backtracking.

Programming Code:

```
% ---------- Facts ----------
fruit(apple).
fruit(banana).
fruit(grapes).
fruit(orange).
fruit(mango).
fruit(strawberry).
fruit(watermelon).

% color(Fruit, Color): a fruit can have more than one color
color(apple, red).
color(apple, green).
color(banana, yellow).
color(grapes, green).
color(grapes, purple).
color(orange, orange).
color(mango, yellow).
color(mango, green).
color(strawberry, red).
color(watermelon, green).

% ---------- Rules ----------
% Prolog backtracks to find every fruit-color pair
fruit_color(F, C) :- fruit(F), color(F, C).

% Two different fruits that have the same color
same_color(F1, F2) :- color(F1, C), color(F2, C), F1 \= F2.

% All colors of a fruit as a list
all_colors(F, List) :- findall(C, color(F, C), List).

% Force backtracking with fail to print every fruit and its color
show_all :-
    fruit_color(F, C),
    format("~w is ~w~n", [F, C]),
    fail.
show_all.
```

OUTPUT:

```
$ swipl -q exp26.pl

?- color(apple, C).
C = red ;
C = green.

?- fruit_color(F, red).
F = apple ;
F = strawberry.

?- fruit_color(mango, C).
C = yellow ;
C = green.

?- same_color(apple, X).
X = strawberry ;
X = grapes ;
X = mango ;
X = watermelon.

?- all_colors(grapes, L).
L = [green,purple].

?- show_all.
apple is red
apple is green
banana is yellow
grapes is green
grapes is purple
orange is orange
mango is yellow
mango is green
strawberry is red
watermelon is green
true.

$
```

Result

The program to find a fruit and its color was implemented successfully using Prolog. Backtracking makes Prolog return every fruit-color combination, one after another, until all the facts have been tried.